Deep magmatic structures of Hawaiian volcanoes, imaged by three-dimensional gravity models

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ABSTRACT

A simplified three-dimensional model for the island of Hawai'i, based on 3300 gravity measurements, provides new insights on magma pathways within the basaltic volcanoes. Gravity anomalies define dense cumulates and intrusions beneath the summits and known rift zones of every volcano. Linear gravity anomalies project southeast from Kohala and Mauna Kea summits and south from Hualālai and Mauna Loa; these presumably express dense cores of previously unrecognized rift zones lacking surface expression. The gravity-modeled dense cores probably define tholeiitic shield–stage structures of the older volcanoes that are now veneered by late alkalic lavas. The three-dimensional gravity method is valuable for characterizing the magmatic systems of basaltic oceanic volcanoes and for defining structures related to landslide and seismic hazards.

Keywords: Hawaii, volcanoes, gravity surveys, magma chambers, rift zones, cumulates.

INTRODUCTION

The island of Hawai'i, the southernmost in the Hawaiian chain, is a collection of five volcanoes and two submarine volcanoes just offshore (Fig. 1); four are considered active (Moore and Clague, 1992). Each of these volcanoes has a central summit vent and at least two rift zones along which eruptions predominantly occur. Each has gone through nearly the same progression in formation: Lō'ihi is the youngest and is in the early tholeiitic shield-building stage, Kīlauea and Mauna Loa are farther along in the same stage, and Hualālai, Mauna Kea, Kohala, and Mahukona are past shield building and into the alkalic capping stage. Understanding how magma rises through these volcanoes from the upper mantle and flows to eruptive vents is fundamental to our understanding of basaltic volcanism and the geologic history of the island of Hawai'i (e.g., Tilling and Dvorak, 1993).

The measurement of gravity field variations is a useful tool for studying the magmatic plumbing system of Hawaiian volcanoes, because magma flows within subsurface conduits, and those conduits are systematically more dense than lava flows. Krivoy and Eaton (1961) and Kinoshita et al. (1963) showed that pronounced positive gravity anomalies are associated with the summits and rift zones of Hawai'i volcanoes, just as they are on O'ahu (Woollard, 1951). These authors proposed that the positive gravity anomalies reflected subsurface dike complexes. Interest in geothermal energy spawned new data acquisition and detailed interpretation (Broyles et al., 1979; Furumoto, 1978) for the East Rift Zone of Kīlauea. These studies modeled the rift zone as a semihorizontal slab 10–40 km wide and 2–3 km thick and having a density of 2700–2900 kg/m^3. Hill and Zucca (1987) summarized joint seismic refraction and gravity data analyses for the island of Hawai'i. Their model requires a massive 2900 kg/m^3 core for Mauna Loa and a 1–2-km-high offshoot beneath Kīlauea. Collectively, these past geophysical studies advanced our understanding of the magmatic structures of active volcanism, but they provided only a partial image of the island-wide complex of magmatic systems. Here we present a three-dimensional interpretation of recently acquired gravity data on the island of Hawai'i that leads to a clearer picture of the entire island's substructure and plumbing systems.

Our approach is to increase the density of gravity measurements over the island and combine them with nearby offshore gravity measurements to represent an area large enough (1400 by 1400 km) so that we can characterize and remove the gravity effects produced by the known downwarping of the oceanic crust beneath the island (e.g., Watts and ten Brink, 1989). The data set is then reduced to a smaller area (250 by 250 km) centered on the island and is interpreted by using a simple, single-layer model representing the most dense rock type thought to occur within the magmatic plumbing—olivine cumulate (3300 kg/m^3). Thus, our model will map the maximum depth to the top of the bodies responsible for the gravity highs on the island and, we predict, will lead to new insights on the plumbing system of an active volcanic island.

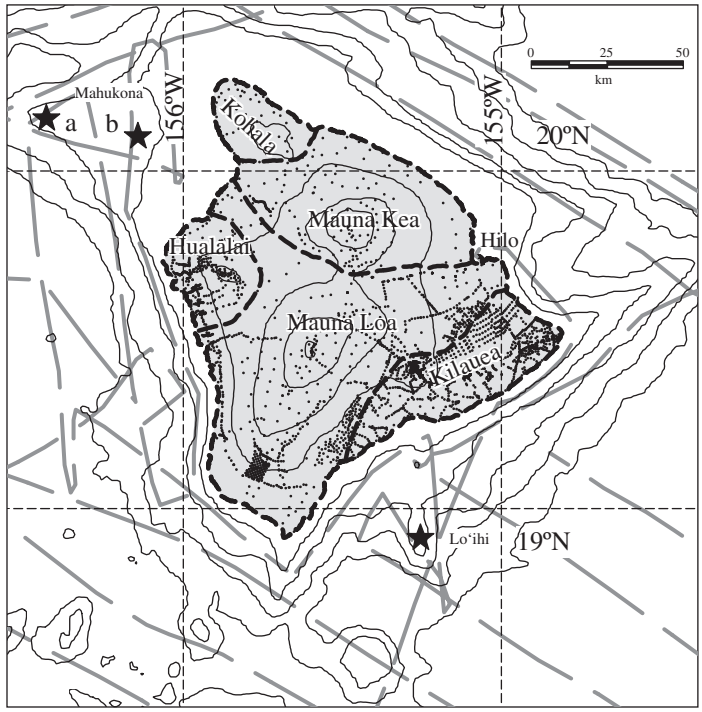

Figure 1. Map of study area showing outline of island of Hawai'i (shaded) and subaerial parts of each of its five volcanoes (short-dash lines). Stars indicate summits of two submarine volcanoes (two for Mahukona and one for Lō'ihi). Garcia et al. (1990) and Clague and Moore (1991) assigned different locations to summit of Mahukona volcano. Dots on island and heavy, long-dash lines offshore (GLORIA shiptracks) are data locations. Thin solid lines are bathymetry and topography contoured at 1 km interval.

DATA DESCRIPTION AND REDUCTION

More than 3300 gravity measurements were made above sea level on the island of Hawai'i (Fig. 1). These data were corrected for instrumental drift and tides, referenced to the ISGN71 benchmark in Hilo, Hawai'i, and reduced to a standard free-air anomaly. The land data were supplemented with marine free-air anomaly data from GLORIA (Ponce et al., 1994) within 200 km of the shoreline and ERS-1, Geosat, and Seasat data (Sandwell and Smith, 1997) farther offshore. Even though our focus was the island, we start with a 1400 km by 1400 km area sufficient to characterize the regional field for proper removal.

The data were corrected for topographic and oceanic terrain effects (Fig. 2). The part of the island edifice above sea level was assigned a density of 2300 kg/m^3, and the part below sea level was assigned a density of 2600 kg/m^3. The reduction densities were chosen on the basis of previous gravity studies, rock-density measurements, and minimization of the gravity expression of topography (Woollard, 1951; Strange et al., 1965; Watts and ten Brink, 1989).

The broad gravity low caused by flexure of oceanic crust beneath all the islands (Watts and ten Brink, 1989) was removed by using a wavelength filter (Hildenbrand et al., 1993). The filter was designed to pass wavelengths shorter than 150 km, partially attenuate components between 150 and 450 km, and completely remove components with wavelengths longer than 450 km. Removing the regional gravity field over the 1400 by 1400 km region reduced the data variance by 99% and left only the short-wavelength gravity highs over the island. The residual corrected gravity field over the island of Hawai'i (Fig. 2) is almost identical in character to the corrected gravity field after removal of ~180 mGal of regional field.

3D INVERSION OF THE RESIDUAL GRAVITY FIELD

Interpretation of the residual gravity data began with 2.5D inversion along four profiles across the summit and rift zones of Kīlauea (Kauahikaua et al., 1996). We found that, in order to satisfy constraints on rift-zone width imposed by gravity gradients and limits on density (should be <3300 kg/m^3, the density of olivine), each interpreted cross section requires a dense core that extends from the mantle through the crust and into the volcanic edifice beneath many eruptive surface features. Each core requires the highest allowable model density of 3300 kg/m^3 at the base (interpreted to be olivine cumulate, Clague and Denlinger, 1994) grading vertically upward into a model density of 2900 kg/m^3 (interpreted to be dense intrusions). Their proximity to surface eruptive features indicates that the dense cores clearly represent the magmatic plumbing that feeds surface eruptive features. We note the interesting result that some of the modeled dense cores are in areas that have no known surface volcanic centers, such as between the Southwest Rift Zones of Kīlauea and Mauna Loa and north of the East Rift Zone of Kīlauea.

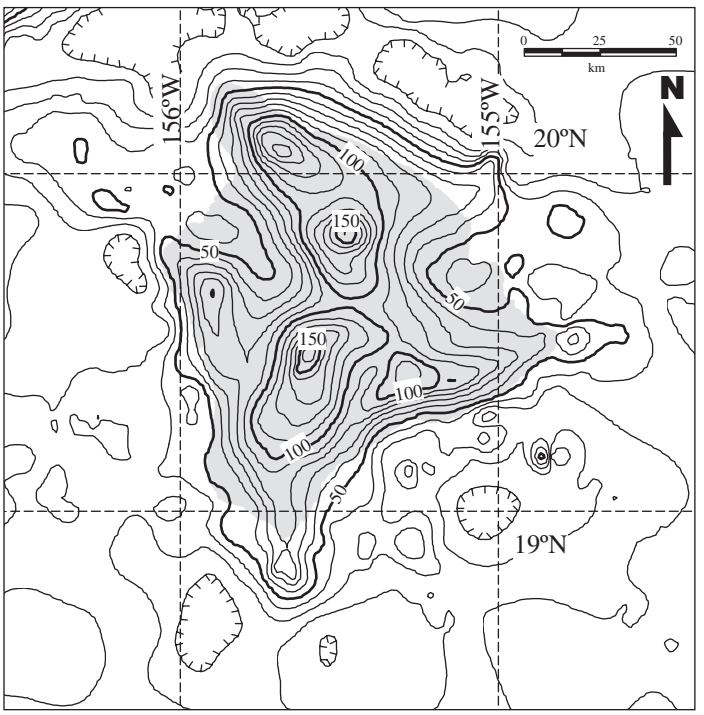

Figure 2. Map of study area showing contoured residual gravity field over island of Hawai'i. Contour interval is 10 mGal. Island is shaded.

To investigate these structures in three dimensions, we use computer program GI3 (Cordell and Henderson, 1968), which is designed to solve for the thickness of a single-density layer of variable thickness with a known upper or lower surface. The model layer consists of a grid of vertical rectangular prisms. The 2.5D-profile interpretations indicate that most of the gravity anomaly is due to the extremely dense rocks, so GI3 was employed to model a layer of 3300 kg/m^3 rocks as if they were the sole source of the anomalous gravity. The 2.5D-profile interpretations also indicate that the cores extend upward from the contact between the crust and the mantle. This situation allows us to fix the lower surface of our model layer there. We selected the mantle depths derived by Watts and ten Brink (1989, their Fig. 12) to represent the crust-mantle contact and our GI3 reference surface (fixed base of the modeled high-density layer). Accordingly, the depth to mantle is fixed at 10.5 km at large distances from the island chain and increases to ~16 km beneath the island. Having defined the density contrast of +700 kg/m^3 (relative to the submarine reduction density of 2600 kg/m^3) and the lower surface of the model layer, GI3 can iteratively build a model up from the reference surface by using a grid of vertical rectangular prisms of that density.

The presence of other mass sources (such as basaltic intrusions or molten magma bodies ~2900 kg/m^3) within the actual magmatic plumbing complex that we have modeled as having only the density of olivine cumulate means that we must view the GI3-interpreted depths to the modeled layer as maxima to the top of the magmatic plumbing. In other words, this interpretation defines the effects of all intrusive activity within each of these volcanoes over their entire history, not just what is active currently. Even though simple, the GI3 model fits the data well and accounts for almost 99% of the variance in the wavelength-filtered, reduced data. The calculated volume of the modeled dense core, which includes olivine cumulate and intrusions, is 31% of the total volume of the island; therefore, the average island density is 2820 kg/m^3.

VOLCANO SUBSTRUCTURE

With the described limitations of the simplified model in mind, we can interpret many features of the GI3 model in terms of volcano structure (Fig. 3). Prominent shallow peaks in the upper model surface are associated with each of the five subaerial volcanoes; more subtle highs are associated with the two submarine volcanoes. The peaks beneath Kohala, Mauna Kea, Mauna Loa, Kīlauea, and Lō'ihi are shallowest beneath their respective topographic summits and presumably represent magma reservoirs and intrusive complexes surrounding them. The dense cores associated with Hualālai and Mahukona (Clague and Moore, 1991; Garcia et al., 1990) are displaced significantly to the south and west of surface features. Linear ridges extending outward from the summits probably reflect intrusive complexes beneath rift zones. The distribution of gravity data is sparsest offshore, so the offshore aspects of the GI3 model are not discussed further. Instead, we concentrate on the volcanologic significance of structures interpreted from gravity data beneath the subaerial volcanoes.

The rift zones of Kīlauea are clearly marked by linear gravity highs (Fig. 3). The Southwest Rift Zone is a minor feature compared to the East Rift Zone and does not extend to the shoreline. The dense central core of this volcano extends substantially south of the summit, suggesting structural continuity between the Southwest Rift Zone and the East Rift Zone, which extends well offshore with an east-northeast trend. This observation supports the single rift zone idea of Fiske and Swanson (1992) for Kīlauea. The surface of the dense core beneath the East Rift Zone dips more steeply to the south than to the north, suggesting southward migration of more recent eruptive activity (Kauahikaua, 1993).

Mauna Loa has two tholeiitic rift zones, a short one extending east-northeast and a long one trending generally south. The short, dense northeast linear substructure in Figure 3 diverges south of eruptive features of Mauna Loa's northeast rift zone toward its distal end. The broad, dense feature extending south from Mauna Loa summit may represent both the current rift zone (to the southwest) and a proto–rift zone (to the south) proposed by Lipman (1980). The latter may have been Mauna Loa's original

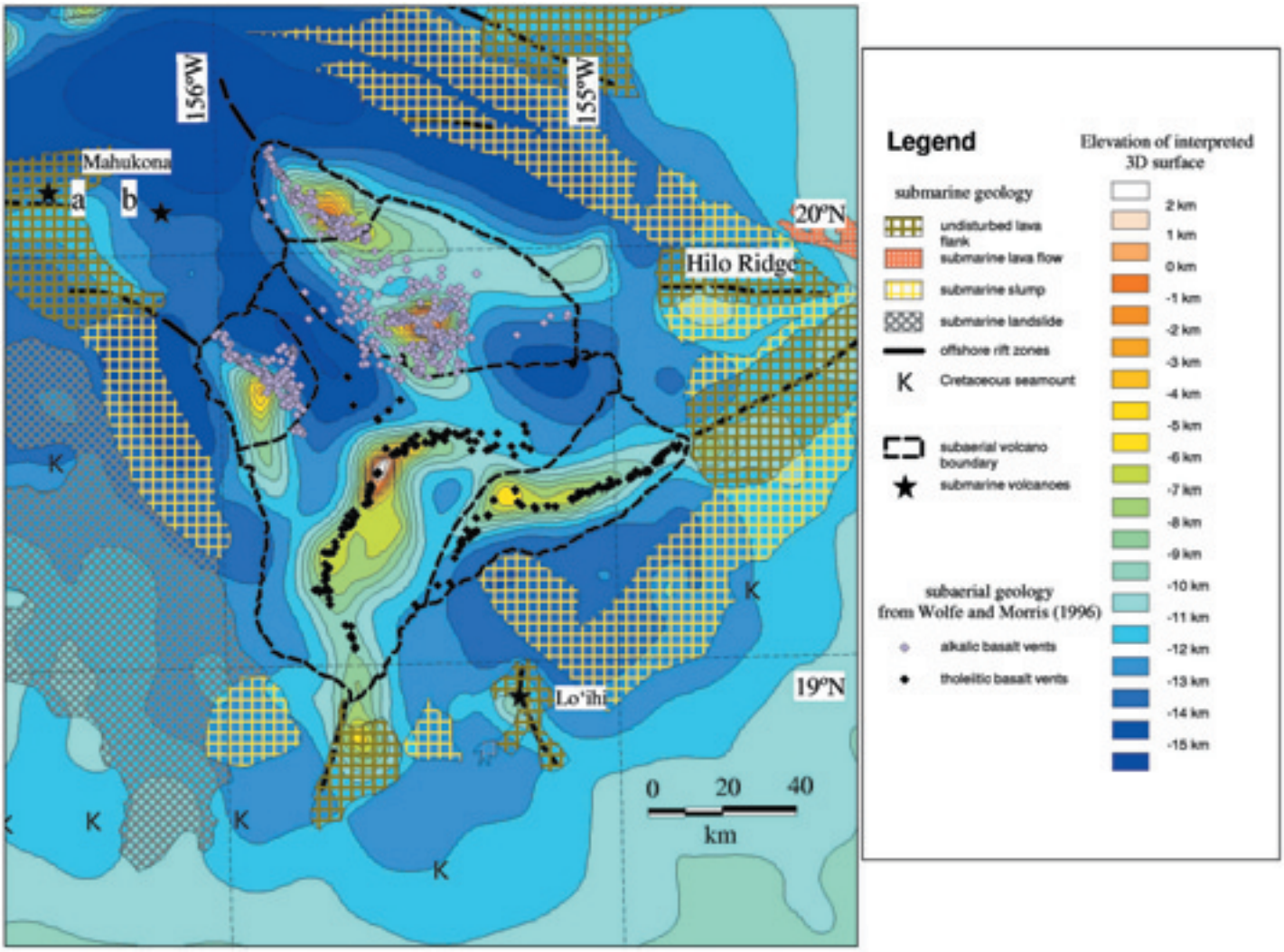

Figure 3. Map of study area showing depth to upper surface of dense volcanic cores interpreted in this study. Contour interval is 1 km. Outlines of island and each of five subaerial volcanoes are dashed. Subaerial geologic features are from Wolfe and Morris (1996) and submarine geologic features are from R.T. Holcomb (1996, written commun.). Rift zones are marked by linear distributions of vents. Alternative locations for summit of Mahukona volcano are shown by a and b.

rift south of the summit, which was abandoned after catastrophic failure of the southwest flank of Mauna Loa (Lipman et al., 1988). As a consequence, eruptive activity may have shifted westward into the head of the debris-flow scar. The present Southwest Rift Zone developed with a westward curve, and its lava flows filled the scar. Its tholeiitic surface features correspond precisely to the underlying dense substructure that extends offshore. The effect of a 40 km length of the proto–south rift zone east of the current rift zone is apparently preserved in the gravity field.

The Northeast and Southwest Rift Zones of Mauna Loa are underlain by dense linear structures (Fig. 3), but there is no indication of dense cores beneath the northwest flank of Mauna Loa, where widely separated radial fissure vents are located (Wolfe and Morris, 1996). The absence of any large volume of dense rocks beneath those vents suggests that they were fed laterally through shallow conduits from the dense cores (i.e., magma storage regions) beneath the summit and upper rift zones.

The dense substructure of Hualālai volcano is displaced at least 4 km south and west of mapped eruptive vents. The surface vents, all of which erupted alkalic basalt, trend northwest and southeast from the topographic summit and do not overlap with the dense linear substructures, which trend northwest and south from a central high. The surface vents are displaced an average of 4 km northeast of the interpreted dense substructures. The northwest-trending linear dense core continues offshore to the submarine northwest rift zone, which is entirely tholeiitic (Clague, 1987). The lack of correlation between dense cores and surface alkalic vents for Hualālai and the correlation between dense cores and the tholeiitic northwest rift zone offshore suggest that the dense cores represent tholeiitic magma pathways.

Mauna Kea has an elliptical-shaped core, slightly elongated east-west, with a broad, linear feature trending southeast. This linear feature may be a buried rift zone of Mauna Kea, although no surface expressions of those rift zones have been mapped. The most recent eruptions of Mauna Kea occurred on the south flank of the volcano (Porter, 1979).

The substructure of Kohala volcano includes a short northwest-trending linear zone and a much longer, subtler east-southeast–trending zone underlying surface eruptive features. The latter rift zone appears to connect almost seamlessly with the submarine feature known as the Hilo Ridge. The Hilo Ridge has been considered to be the offshore extension of the East Rift Zone of Mauna Kea; however, our results suggest that the Hilo Ridge substructure is related to Kohala. Holcomb et al. (2000) came to the same conclusion, based on (1) discovery of a submarine terrace on Hilo Ridge that may correlate with one on Kohala but that is absent from Mauna Kea and (2) a better match between isotopic compositions of Hilo Ridge and Kohala basalt than those involving Mauna Kea.

INTERVOLCANO STRUCTURES

Above a depth of 10 km below sea level, the dense cores of each volcano are distinct and are not connected. Possible connections between volcanoes are visible at depths from 10 to 14 km. The dense cores representing the Northwest Rift Zone and summit of Kohala, the Southeast Rift Zone of Mauna Kea, and the summit of Kīlauea appear to connect at depth. In addition, the South Rift Zone of Hualālai and the southern part of the Southwest Rift Zone of Mauna Loa could be connected. The first of these is similar to the northernmost loci of Hawaiian volcanoes first pointed out by Dana (1849); however, his southernmost loci aligning Hualālai, Mauna Loa, and Kīlauea are not reflected in the distribution of dense cores.

The connections between volcanoes in Figure 3 are at a depth that would place them partially within the oceanic crust and suggest that magma may have exploited linear zones of weakness or cracks in the oceanic plate while initiating island building and before centralizing into separate conduits. An alternative hypothesis is that island-building volcanoes can originate along the rift zones of older volcanoes. For example, Kīlauea may have developed along a southeast rift zone of Mauna Kea before its magma supply system became independent.

DISCUSSION AND CONCLUSIONS

Our interpretation of gravity data on and around the island of Hawai'i has yielded an unprecedented view of the substructure of each volcano making up the island and helps lead to a greater understanding of the development of magma pathways during island building. Interpreted dense cores directly underlie surface rift zone and summit features of the tholeiitic volcanoes Kīlauea and Mauna Loa. They also underlie most, but not all, surface eruptive features of the older volcanoes Mauna Kea and Kohala, which are capped by alkalic lava flows. The interpreted dense cores are significantly displaced south and west of the surface eruptive features of Hualālai.

Their placement is consistent with our interpretation of the cores as elements of the tholeiitic magma plumbing system, in the form of high-density olivine cumulates grading upward into dikes and intrusions beneath the island's volcanic summits and rift zones. Tholeiitic magma chambers have the shape of narrow vertical cones or cylinders and rift zones as vertical walls of dense material extending up from the base of the crust. These components of the tholeiitic magma pathways must rise within the tholeiitic shield as the volcano grows (Clague, 1987); the most recently active part is on top, and a trail of older, solidified pathways is below. Alkalic lavas that cap the tholeiitic shields are much less abundant and their pathways are apparently not massive enough to be detectable by using gravity mapping.

Deep linear alignments between Kohala, Mauna Kea, and Kīlauea, and between Hualālai and the Southwest Rift Zone of Mauna Loa, suggest either that magma may have exploited cracks in the oceanic plate before centralizing into separate vents or that volcanoes can develop along the rift zones of older volcanoes. Pertinent future studies using the three-dimensional model applied here may lead to a better understanding of the role of the identified structures and boundaries in the release of seismic energy (e.g., rift margins) and in processes leading to giant landslides (e.g., geometry of the deeper parts of the island).

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